

PHYSICS LEARNING IN VIDEO MODELS

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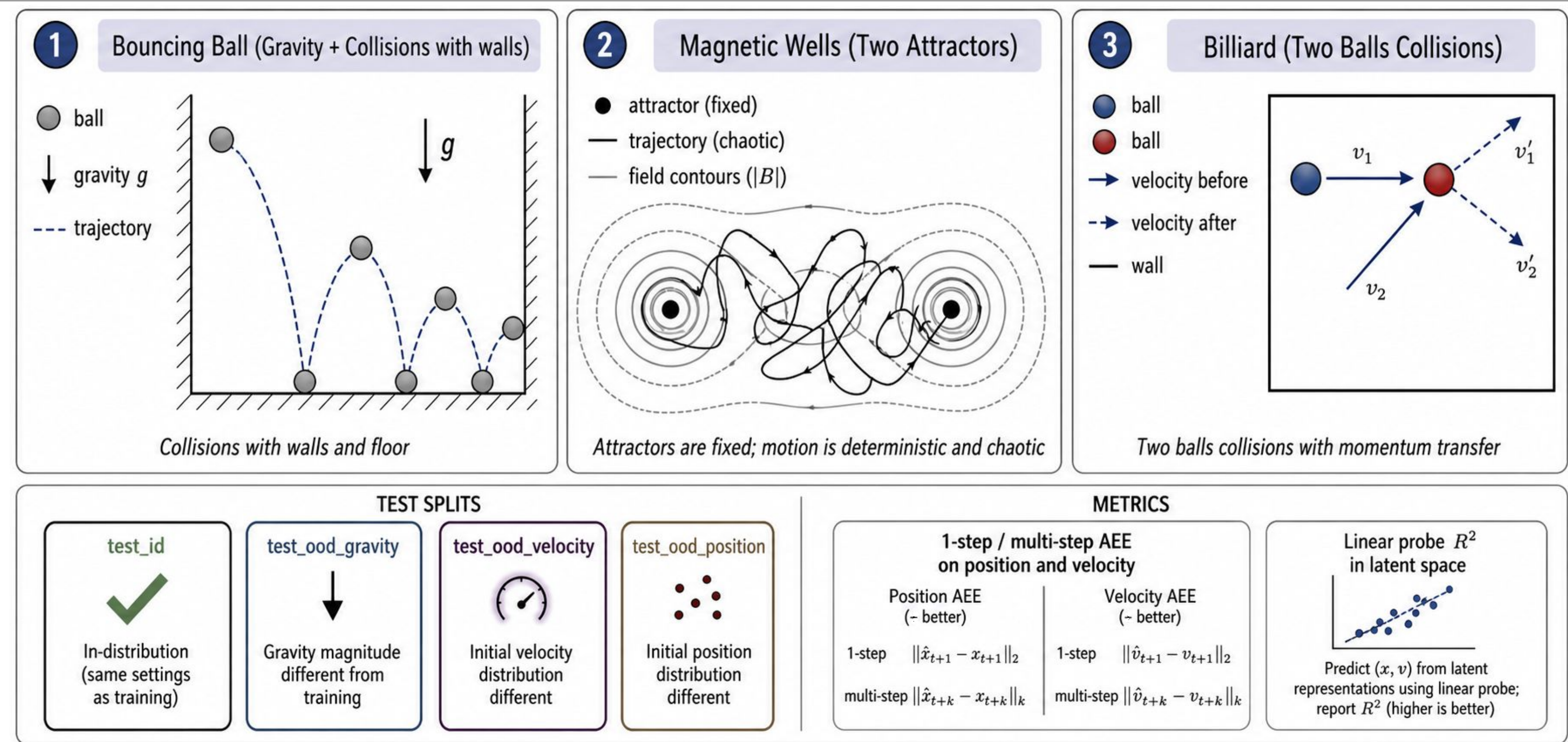
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Introduction

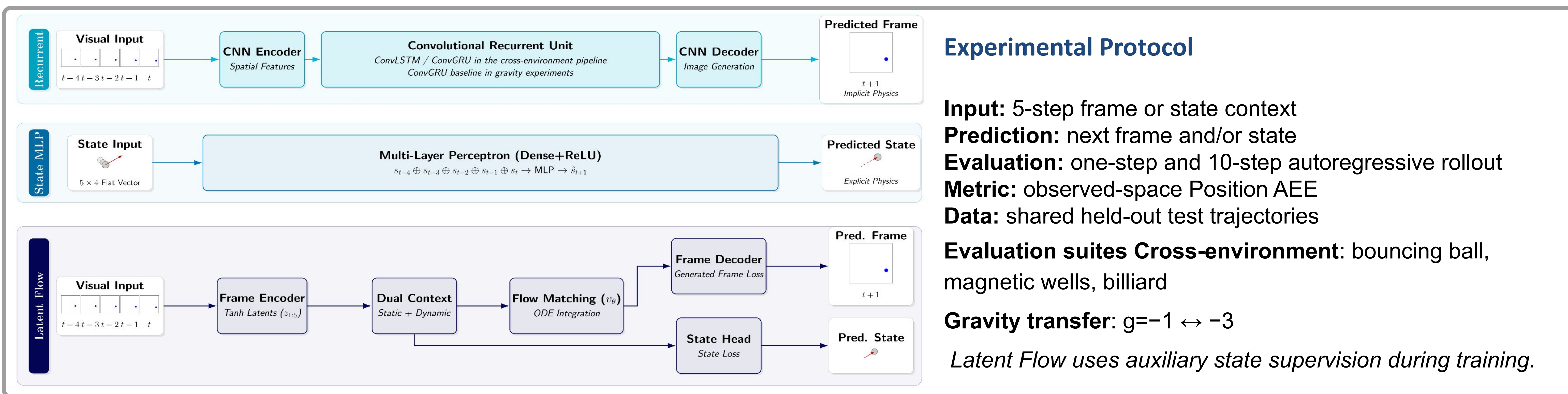
Problem: Do generative video models truly grasp physical laws (e.g., gravity, momentum) or merely replicate visual patterns?

Core Research Questions:

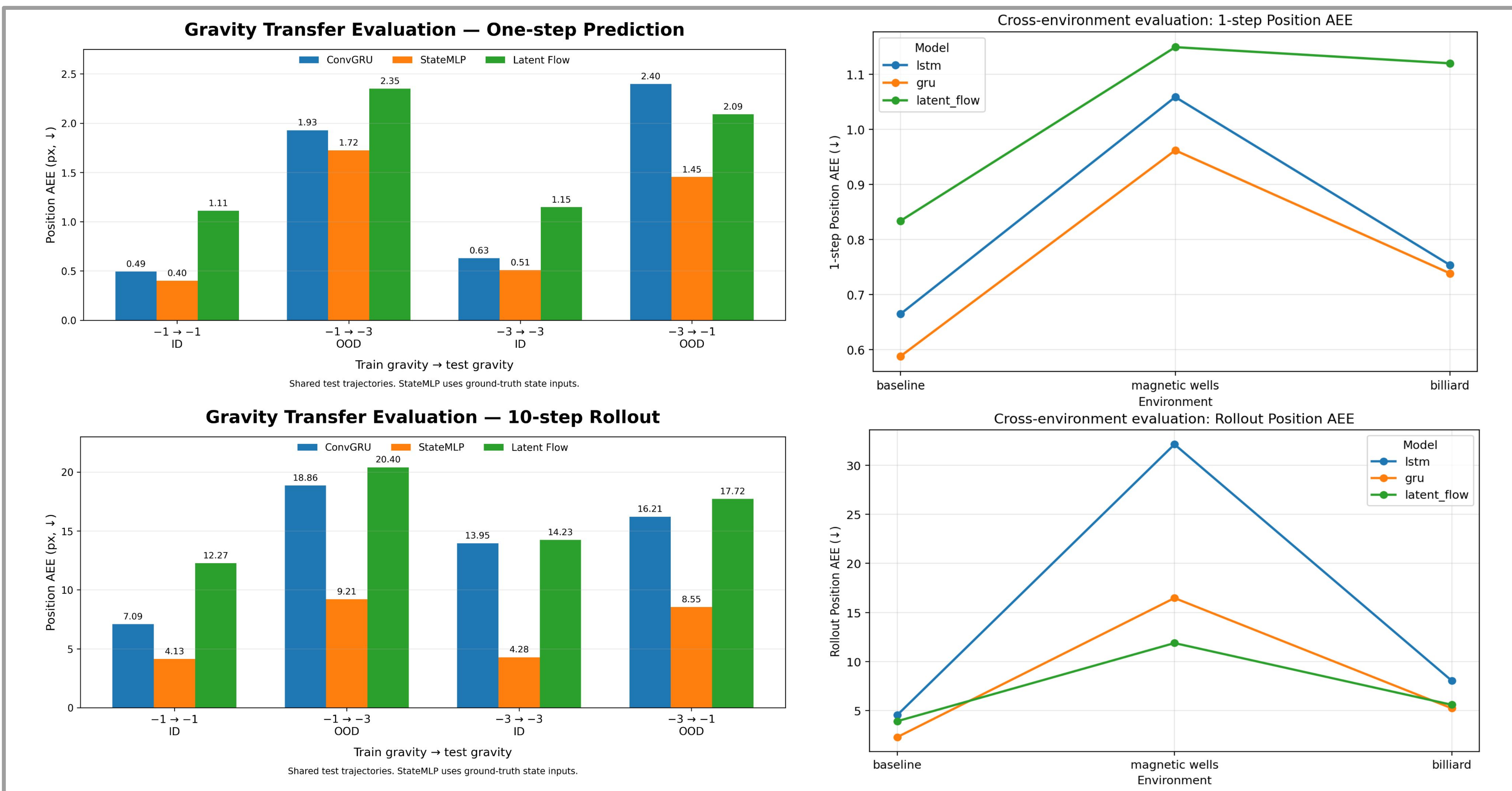
- How do generative architectures (Latent Flow Matching) compare to recurrent baselines (LSTM/GRU) at learning physical dynamics?
- Does an explicit state-based model (trained on ground-truth coordinates) generalize better to unseen scenarios than implicit visual/latent models?
- How robust are these models under out-of-distribution physical shifts (e.g., altered gravity, unseen velocities, or spawn distributions)?



Method



Quantitative Results



[1] Physics learning in Bouncing Ball environment: <https://github.com/eliaskempf/world-model-physics>

[2] Orbis Flow Matching Video model: <https://github.com/lmb-freiburg/orbis>

[3] ByteDance paper studying studying physics learning in video models: <https://arxiv.org/pdf/2411.02385>